

1.1–1.2 Dalton's Atomic Theory and the Modern Atomic Theory

1. The Greek philosopher Democritus proposed that matter consists of tiny, indestructible particles. He called these particles “atomos”, a word meaning

(a) invisible (b) indivisible (c) indestructible (d) infinite

3. Regardless of its source, 1.00 g of water decomposes into 0.112 g of hydrogen and 0.888 g of oxygen. This illustrates the law of

(a) conservation of energy (b) definite (constant) proportions (c) multiple proportions (d) conservation of mass

5. Which of the following postulates of Dalton's atomic theory continues to be accepted (is retained) in the modern atomic theory?

(a) All atoms of an element are identical in mass (b) Atoms are indivisible (c) Matter is continuous and infinitely divisible (d) Atoms combine in small whole-number ratios to form compounds

2. John Dalton's atomic theory, which gained broad acceptance nearly 2000 years after Democritus, was proposed in

(a) 1808 (b) 1789 (c) 1897 (d) 1911

4. In two compounds of nitrogen and oxygen, 1.750 g of nitrogen combines with 1 g of oxygen in compound A, while 0.4375 g of nitrogen combines with 1 g of oxygen in compound C. The ratio of the masses of nitrogen (A : C) is

(a) 1 : 4 (b) 2 : 1 (c) 4 : 1 (d) 3 : 2

6. Because of the discovery of isotopes, the modern atomic theory modified Dalton's postulate that

(a) atoms are indivisible (b) all atoms of the same element are identical in every respect (c) atoms are destroyed during chemical reactions (d) matter is continuous

1.3.1 The Discovery of the Electron

7. The modern theory of the physical (internal) structure of the atom is generally said to have begun when J.J. Thomson discovered the electron in

(a) 1873 (b) 1897 (c) 1909 (d) 1911

9. A beam of cathode rays bends away from a negatively charged plate and toward a positively charged plate. Thomson concluded that cathode rays

(a) are beams of negatively charged particles (b) are beams of positively charged particles (c) consist of uncharged radiation (d) are streams of light

11. By measuring the deflection of a cathode-ray beam in electric and magnetic fields of known strengths, Thomson was able to calculate

(a) the charge of the electron alone (b) the mass of the electron alone (c) the speed of the electrons (d) the ratio of the electron's mass to its charge

8. In Thomson's cathode-ray tube, the electrode from which cathode rays originate — the negative electrode — is called the

(a) anode (b) filament (c) cathode (d) grid

10. Thomson found that the properties of cathode rays were the same regardless of the material of which the cathode was made. He concluded that

(a) electrons are found only in the cathode (b) electrons are constituents of all matter (c) cathode rays travel from the anode to the cathode (d) the charge on an electron is positive

12. In 1909, Robert Millikan's oil-drop experiment measured

(a) the mass of the neutron (b) the speed of light (c) the charge-to-mass ratio of the proton (d) the charge on an electron, $-1.602 \times 10^{-19} \text{ C}$

1.3.2 Radioactivity and the Discovery of the Nucleus

13. Radioactivity (radioactive decay) is best defined as the

- (a) spontaneous emission of particles and/or radiation from the unstable nuclei of certain atoms
- (b) absorption of radiation by stable nuclei
- (c) controlled splitting of heavy nuclei by neutrons
- (d) emission of X-rays from hot metal filaments

15. Beta (β) particles, emitted by radioactive substances, are

- (a) helium nuclei
- (b) protons released from the nucleus
- (c) uncharged rays of high energy
- (d) electrons coming from inside the nucleus

17. Thomson's "plum-pudding" model described the atom as

- (a) a tiny dense nucleus surrounded by orbiting electrons
- (b) electrons and protons randomly distributed in a positively charged cloud
- (c) empty space containing a massive nucleus
- (d) a solid sphere containing no electric charges

1.3.3–1.4 The Neutron and the Subatomic Particles

19. In 1932, James Chadwick discovered the neutron while studying the penetrating radiation given off when the metal ____ is irradiated with alpha rays

- (a) gold
- (b) beryllium
- (c) nitrogen
- (d) aluminum

21. The mass of a neutron (1.67493×10^{-27} kg) is about how many times the mass of an electron?

- (a) About 1840
- (b) About 184
- (c) About 2
- (d) About 18,400

23. The atomic number (Z) of an element equals the number of

- (a) protons in the nucleus of each of its atoms
- (b) neutrons in its nucleus
- (c) protons plus neutrons in its nucleus
- (d) electrons in its outermost shell

1.4.2 Atomic Mass and Isotopes

25. Isotopes of an element are atoms that

- (a) have the same mass number but different atomic numbers
- (b) have the same number of neutrons but different numbers of protons
- (c) have the same atomic number but different numbers of

14. An alpha (α) particle is best described as

- (a) a helium nucleus
- (b) a high-speed electron
- (c) a neutron
- (d) a hydrogen ion

16. Which statement is true of gamma (γ) rays?

- (a) They are positively charged and deflected by a negative plate
- (b) They are identical in nature to beta particles
- (c) They are attracted by a positively charged plate
- (d) They carry no charge and are not deflected by electric or magnetic fields

18. The main result of Rutherford's gold-foil experiment was that the atom is

- (a) mostly empty space containing a tiny, massive, positively charged nucleus
- (b) a solid, massive sphere
- (c) a uniform positive cloud with embedded electrons
- (d) a large diffuse nucleus filling the whole atom

20. The neutron is a nuclear particle having

- (a) a mass about equal to that of an electron and a negative charge
- (b) a mass about equal to that of a proton and a positive charge
- (c) a mass about twice that of a proton and no charge
- (d) a mass almost identical to that of the proton and no electric charge

22. One atomic mass unit (amu) is defined as

- (a) 1/16 the mass of an oxygen-16 atom
- (b) 1/12 the mass of an atom of carbon-12
- (c) the mass of one neutron
- (d) the mass of one electron

24. The mass number (A) of an atom is equal to the number of

- (a) protons plus electrons
- (b) protons plus neutrons in the nucleus
- (c) neutrons only
- (d) protons only

26. How many protons and neutrons are there in an atom of aluminum-27 ($^{27}_{13}\text{Al}$)?

- (a) 14 protons and 13 neutrons
- (b) 27 protons and 13 neutrons
- (c) 13 protons and 14 neutrons
- (d) 13 protons and 27 neutrons

neutrons (d) always differ in the number of electrons

27. An atom of lead-207 ($^{207}_{82}\text{Pb}$) contains

- (a) 82 protons and 207 neutrons (b) 82 protons and 125 neutrons (c) 125 protons and 82 neutrons (d) 207 protons and 82 neutrons

29. The atomic mass of an element that occurs as a mixture of isotopes is the

- (a) mass of its heaviest isotope (b) mass of its most abundant isotope (c) unweighted average of all its isotopic masses (d) average of the isotopic masses weighted by their fractional abundances

31. Element X has four naturally occurring isotopes that all contain 24 protons; the isotopes contain 26, 28, 29, and 30 neutrons. Element X is

- (a) manganese (b) iron (c) chromium (d) vanadium

1.5.1 Electromagnetic Radiation

33. Electromagnetic waves travel through a vacuum with a speed (c) of

- (a) 3.0×10^8 m/s (b) 3.0×10^6 m/s (c) 3.0×10^{10} m/s (d) 3.0×10^5 m/s

35. Frequency (ν) is the number of cycles a wave undergoes per second; its unit is the

- (a) joule (b) hertz (Hz) (c) meter (d) newton

37. The visible spectrum extends from violet ($\lambda = 380$ nm) to red ($\lambda = 750$ nm). Which color corresponds to the longest wavelength and therefore the lowest frequency?

- (a) Blue (b) Yellow (c) Red (d) Violet

1.5.2 The Quantum Theory and the Photon

39. Max Planck (1900) proposed that atoms and molecules can emit or absorb energy only in discrete quantities — in small packets called

- (a) molecules (b) isotopes (c) nucleons (d) quanta

28. Naturally occurring carbon is 98.89% carbon-12, 1.11% carbon-13, and a trace of carbon-14. The most abundant isotope of carbon is

- (a) carbon-13 (b) carbon-14 (c) carbon-12 (d) all three in equal amounts

30. Silver is 51.84% ^{107}Ag (106.90509 amu) and 48.16% ^{109}Ag (108.90476 amu). The atomic mass of silver is

- (a) 106.91 amu (b) 107.87 amu (c) 108.90 amu (d) 107.00 amu

32. Boron has two natural isotopes, ^{10}B (10.013 amu) and ^{11}B (11.009 amu), and its atomic mass is 10.811 amu. Which isotope occurs in the greater abundance

- (a) ^{10}B (b) both isotopes in equal amounts (c) ^{11}B (d) no definite conclusion is possible

34. The wavelength (λ) of a wave is the

- (a) number of cycles the wave undergoes per second (b) time taken for one complete cycle (c) distance the wave travels during one cycle (d) maximum height of the wave above its axis

36. Because $c = \lambda\nu$, for electromagnetic radiation traveling at a fixed speed, the frequency and the wavelength are

- (a) directly proportional (b) inversely proportional (c) equal to each other in value (d) independent of each other

38. Ethiopian National Radio (Addis Ababa) broadcasts an AM signal at a frequency of 2400 kHz. Given $c = 3.00 \times 10^8$ m/s, the wavelength of these radio waves is

- (a) 1.25 m (b) 12.5 m (c) 1250 m (d) 125 m

40. Planck's constant (h) has the value

- (a) 6.63×10^{-34} J·s (b) 3.00×10^8 m/s (c) 2.18×10^{-18} J (d) 1.602×10^{-19} C

41. The energy (E) of a single quantum of radiation of frequency ν is given by

- (a) $E = h\nu$ (b) $E = mc^2$ (c) $E = h\nu/\lambda$ (d) $E = v/h$

43. The photoelectric effect is the ejection of electrons from a metal surface by light; the minimum frequency of light that causes electron ejection is called the

- (a) resonance frequency (b) threshold frequency, ν_0
(c) Lyman frequency (d) Planck frequency

42. According to quantum theory, a system can transfer energy only

- (a) in any arbitrarily small amount (b) in whole quanta (integral multiples of $h\nu$) (c) in amounts smaller than $h\nu$ (d) in amounts of exactly $2h\nu$

44. When light of frequency ν ejects an electron from a metal with threshold frequency ν_0 , the kinetic energy (KE) of the ejected electron is

- (a) $KE = h\nu + h\nu_0$ (b) $KE = h\nu_0 - h\nu$ (c) $KE = h\nu - h\nu_0$ (d) $KE = mc^2$

1.5.3–1.5.5 Atomic Spectra and the Bohr Model of Hydrogen

45. When an electric discharge passes through hydrogen gas and the emitted light is viewed through a prism, the spectrum observed is

- (a) a series of discrete lines characteristic of hydrogen (b) a continuous rainbow like that of sunlight (c) a single broad colored line (d) a pattern identical for every element

47. According to the Bohr model of the hydrogen atom, the electron

- (a) can occupy an orbit of any radius (b) moves in a circular orbit, and only certain orbits are allowed (c) is randomly distributed in a positively charged cloud (d) radiates energy continuously while orbiting

49. Bohr showed that the radius of the n th orbit of the hydrogen electron is $r = (0.53 \text{ \AA})n^2$. The radius of the orbit with $n = 2$ is

- (a) $0.53 \text{ \AA}$ (b) $2.12 \text{ \AA}$ (c) $1.06 \text{ \AA}$ (d) $4.24 \text{ \AA}$

51. When the hydrogen electron falls from $n = 3$ to $n = 2$, an energy of $3.03 \times 10^{-19} \text{ J}$ is released as a photon. The wavelength of this photon is

- (a) 486 nm (b) 656 nm (c) 122 nm (d) 1220 nm

46. An atom emits light when an electron

- (a) absorbs energy and jumps to a higher energy state (b) stays within the same orbit (c) is completely removed from the atom (d) returns from a higher energy state to a lower energy state

48. Bohr's quantum condition states that the angular momentum of the electron (mvr) must be

- (a) an integral multiple of $h/2\pi$ (b) always equal to h (c) proportional to the square of the radius (d) equal to the electron's kinetic energy

50. Using $E_n = -2.18 \times 10^{-18} \frac{\text{J}}{n^2}$, the ground state of the hydrogen electron — the lowest-energy, most stable state — has

- (a) $n = 1$ (b) $n = 2$ (c) $n = 3$ (d) $n = \infty$

52. Which of the following is a limitation of the Bohr model of the atom?

- (a) It does not explain the spectra of atoms with more than one electron, such as helium (b) It explains the spectrum of every atom completely (c) It cannot account for the hydrogen line spectrum (d) It works only in the presence of a magnetic field

1.5.6–1.7 The Quantum Mechanical Model, Configurations, and the Periodic Table

53. According to de Broglie (1924), the wavelength of a particle of mass m moving with speed v is

- (a) $\lambda = h/mv$ (b) $\lambda = hmv$ (c) $\lambda = mv/h$ (d) $\lambda = mh/v$

54. Using $\lambda = h/(mv)$, the de Broglie wavelength of an electron ($m = 9.11 \times 10^{-31} \text{ kg}$, $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$) moving at $1.00 \times 10^6 \text{ m/s}$ is approximately

55. Heisenberg's uncertainty principle states that it is impossible to determine simultaneously, with precision

(a) the mass and the charge of an electron (b) the position and the momentum of an electron (c) the energy and frequency of light (d) the speed and wavelength of light

57. On the Pauling scale, a value of 4.0 is assigned to the element with the greatest ability to attract bonding electrons. The most electronegative element is

(a) chlorine (b) oxygen (c) fluorine (d) nitrogen

59. Pauli's exclusion principle states that

(a) orbitals are each occupied by a single electron before pairing begins (b) no two electrons in an atom can have the same set of four quantum numbers (c) electrons occupy the lowest-energy orbitals first (d) a single orbital holds two electrons of the same spin

(a) 7.3 nm (b) 73 nm (c) 0.073 mm (d) 0.73 nm

56. In quantum mechanics, the quantity ψ^2 — the square of the wave function — gives the

(a) probability of finding the electron in a given region of space (b) exact trajectory of the electron (c) speed of the electron (d) total energy of the atom

58. Which of the following orbital designations does NOT exist in an atom?

(a) 1s (b) 2d (c) 3p (d) 4f

60. The observed ground-state electron configuration of chromium ($Z = 24$) is

(a) $[\text{Ar}]4s^13d^5$ (b) $[\text{Ar}]4s^23d^4$ (c) $[\text{Ar}]4s^13d^4$ (d) $[\text{Ar}]4s^23d^5$